GREEAT

GRaphene to Electrons: Energy and Angular resolved Transmission

Proposal for Young Researcher Grant CSN5 26724/2024

Proponent: Alice Apponi

Participating units: - INFN Sezione di Roma Tre

- Dipartimento di Scienze,

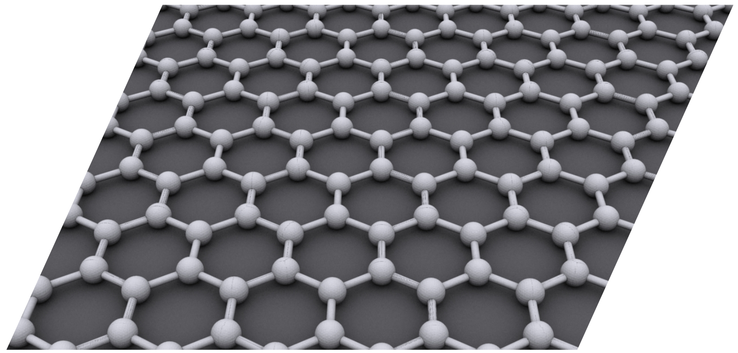
Università degli studi Roma Tre

Presidenza INFN - 19.11.2024

Graphene Transparency: a Growing Topic of Interest

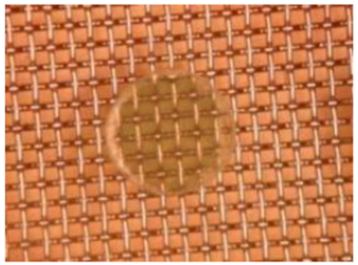
Graphene:

- Single sheet 1 atom thick
- C atoms arranged in honeycomb lattice

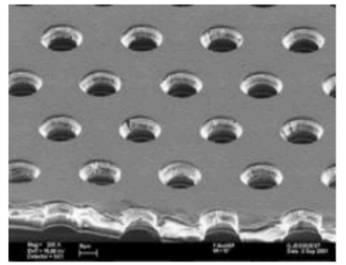


Integration of graphene in MPGD:

- Transparency to electrons
- Impermeability to atoms



MicroMegas



GEM

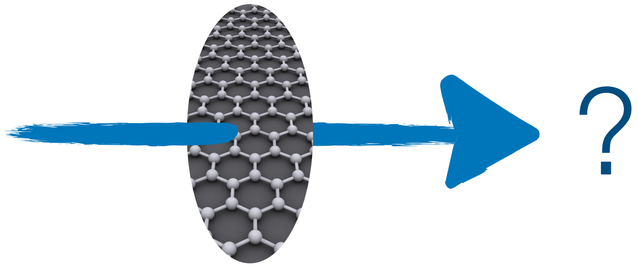
PTOLEMY experiment:

- Neutrino mass
- Tritiated graphene target

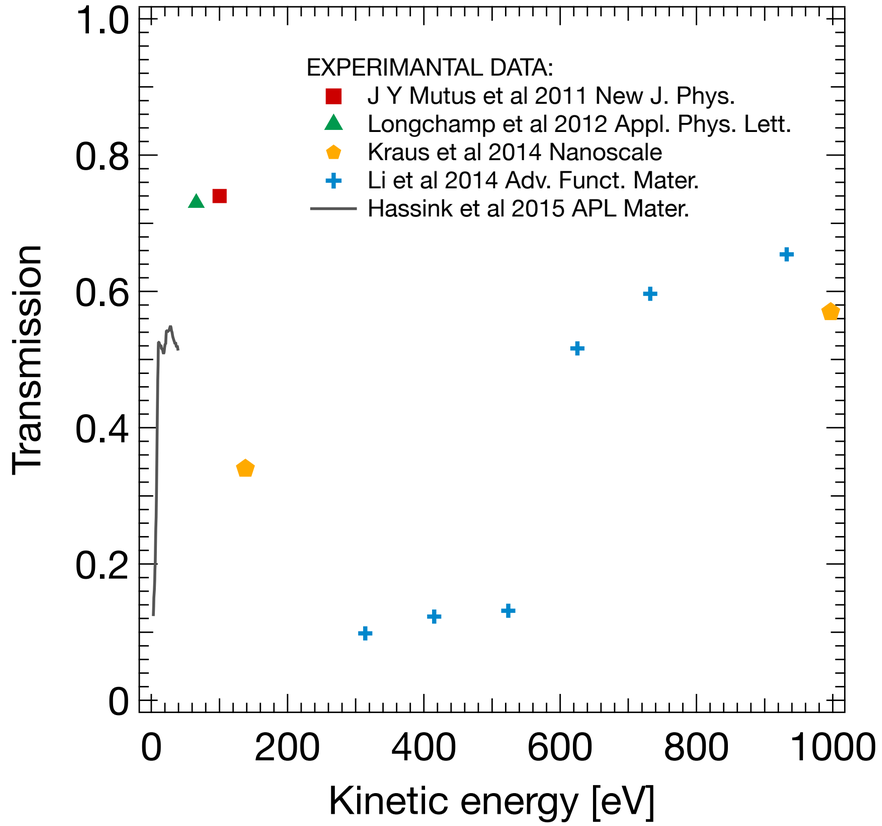
Graphene Transparency: Lots of Interest But Lack of Information

Transmission of low-energy electrons through graphene:

- Only a few data below 1 keV
- Clear lack of information
- Cross-section: still unknown



Quality? Cleannes?



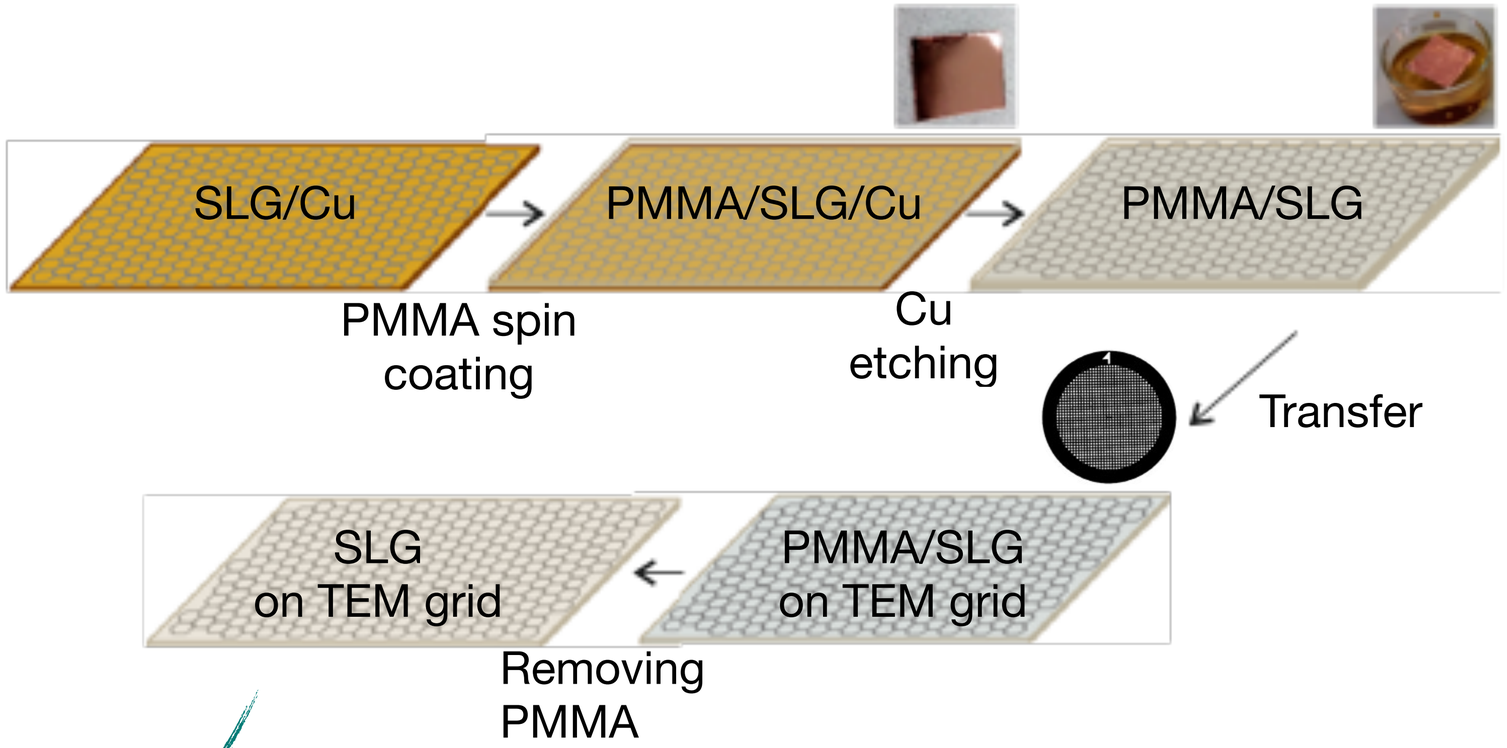
My PhD thesis included:

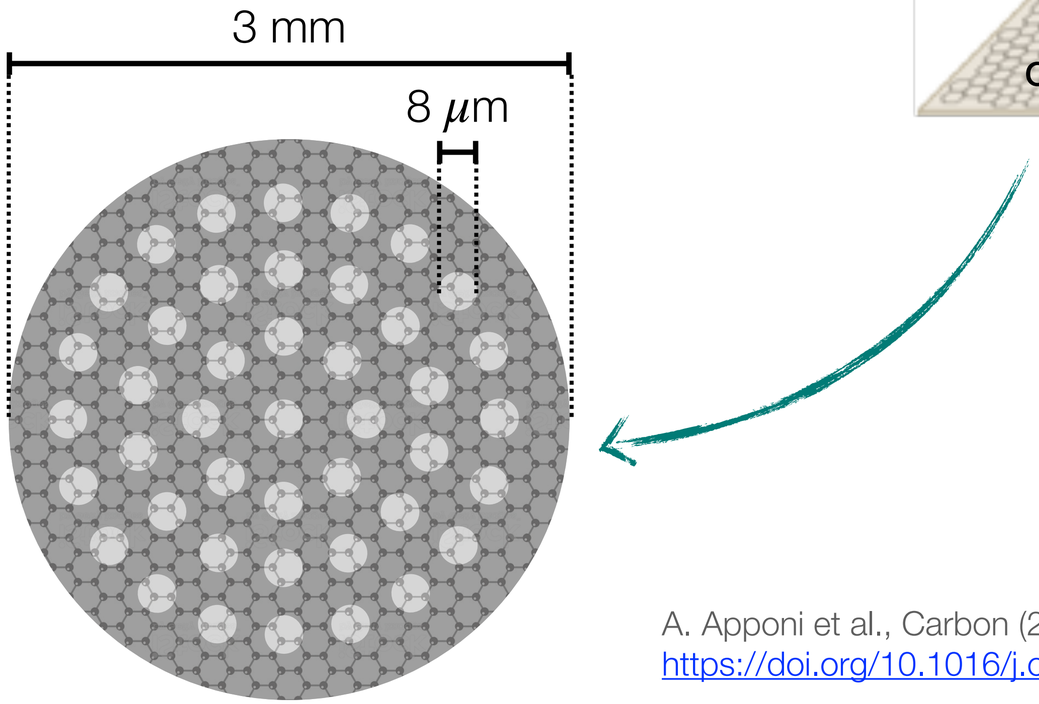
- Graphene thorough characterisation
- Transmission measurement in an extended energy range (30 - 900 eV)

Our Samples: Graphene Transferred Onto Metallic Grids

Monolayer graphene on nickel grids:

- Graphene grown on copper
- Transfer onto commercial nickel grids (Ted Pella Inc. - G2000HAN)
- Access to suspended graphene regions



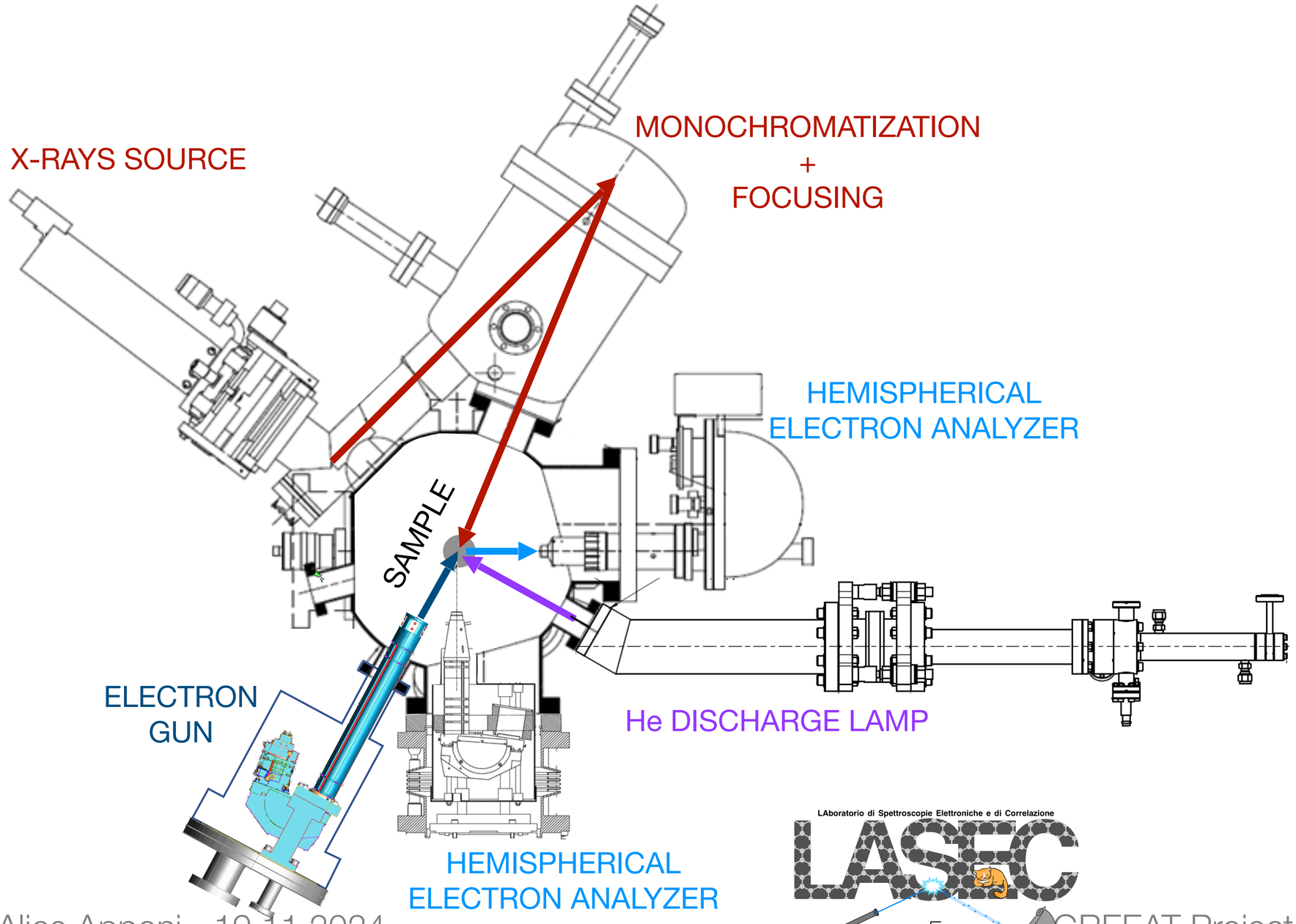




A. Apponi et al., Carbon (2024), https://doi.org/10.1016/j.carbon.2023.118502

UTMOST
UltraThin Materials fOr gaSeous deTectors

LASEC Lab: Electronic Spoctroscopies and More…



X-ray Al Kα source:

- hv = 1486.7 eV
- Monochromatized beam
- XPS resolution = 0.46 eV

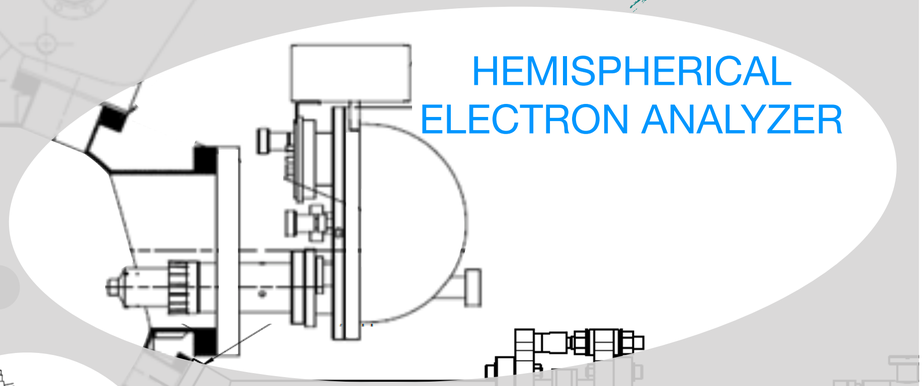
Custom-made monochromatic electron gun:

- Continuous electron beam
- Tuneable energy 30 - 900 eV
- Resolution = 45 meV

He discharge lamp:

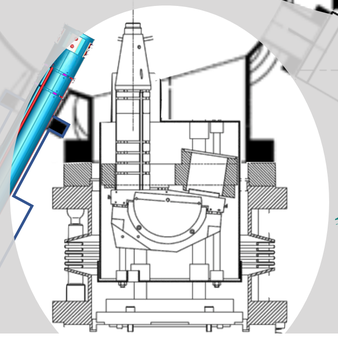
- Spot diameter 300 μm

LASEC Lab: Electronic Spoctroscopies and More…



HEMISPHERICAL
ELECTRON ANALYZER

Equipped with MCP + delay line
for parallel acquisition



HEMISPHERICAL
ELECTRON ANALYZER

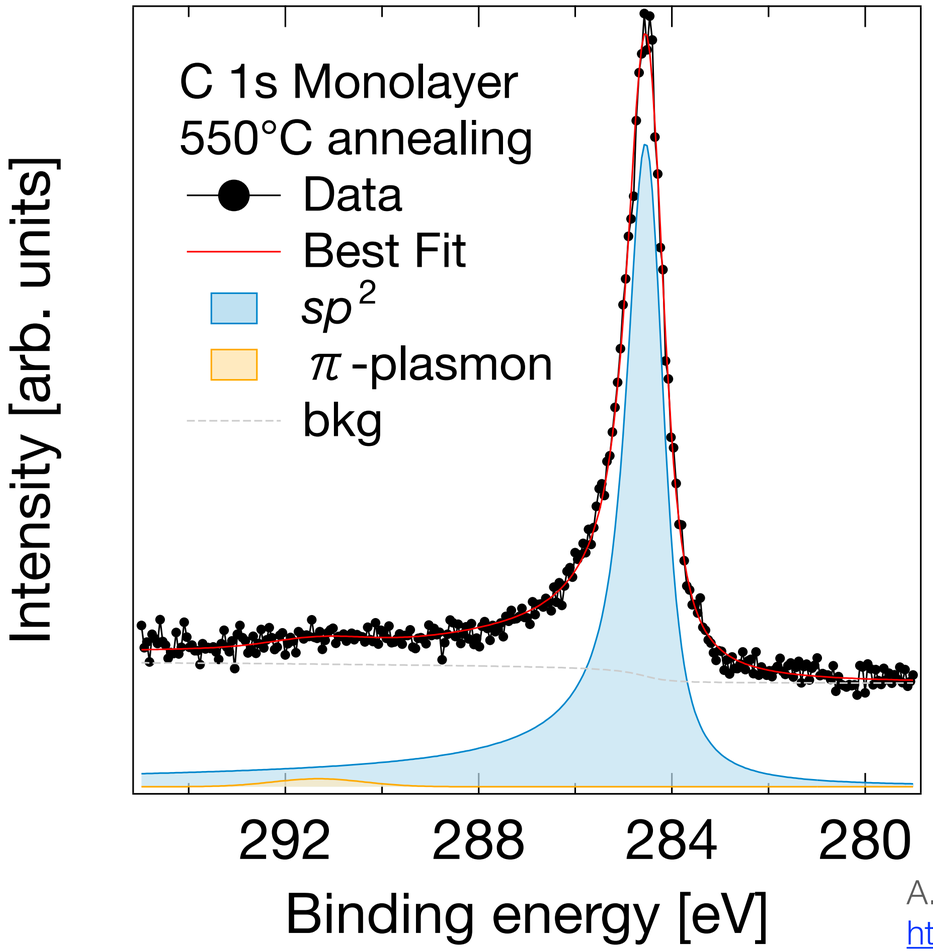
"Twin" analayzer of the one we
will use in the new chamber

LAboratorio di Spettroscopie Elettroniche e di Correlazione

Spectroscopic Characterisation Reveals High Quality Graphene

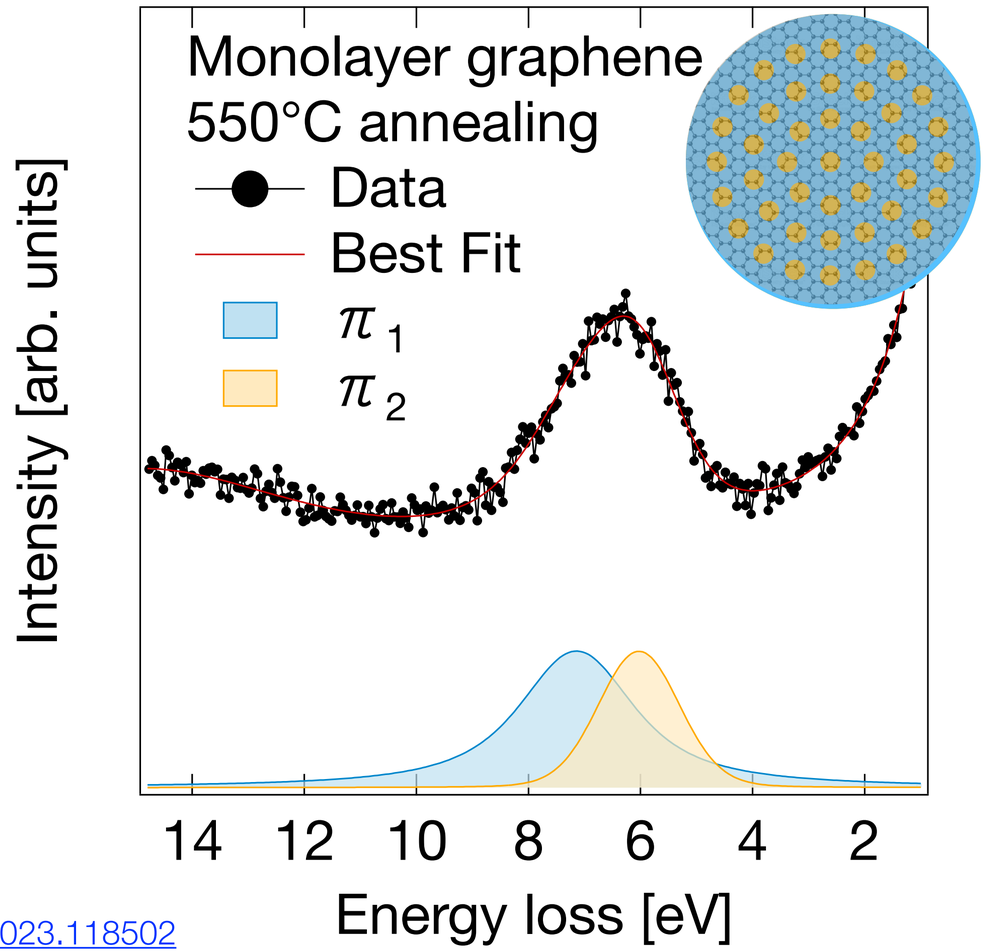
X-ray photoemission spectroscopy (XPS):

- C 1s core-level spectrum
- Pure sp^2 contribution —> **high quality** + **cleannes!**



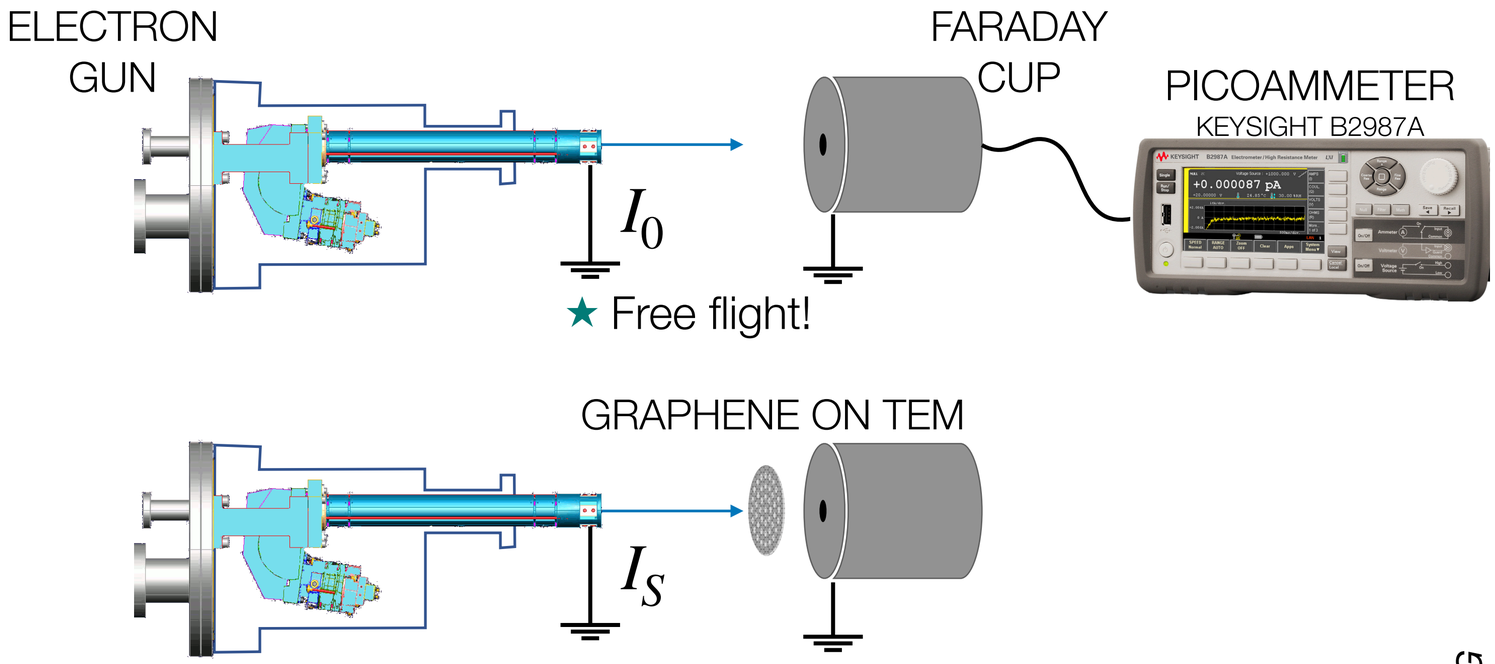
Electron energy loss spectroscopy (EELS):

- π-plasmon excitation
- **High quality** + suspended footprint



A. Apponi et al., Carbon (2024), https://doi.org/10.1016/j.carbon.2023.118502

Graphene Transmission Measured in the 30 - 900 eV Range

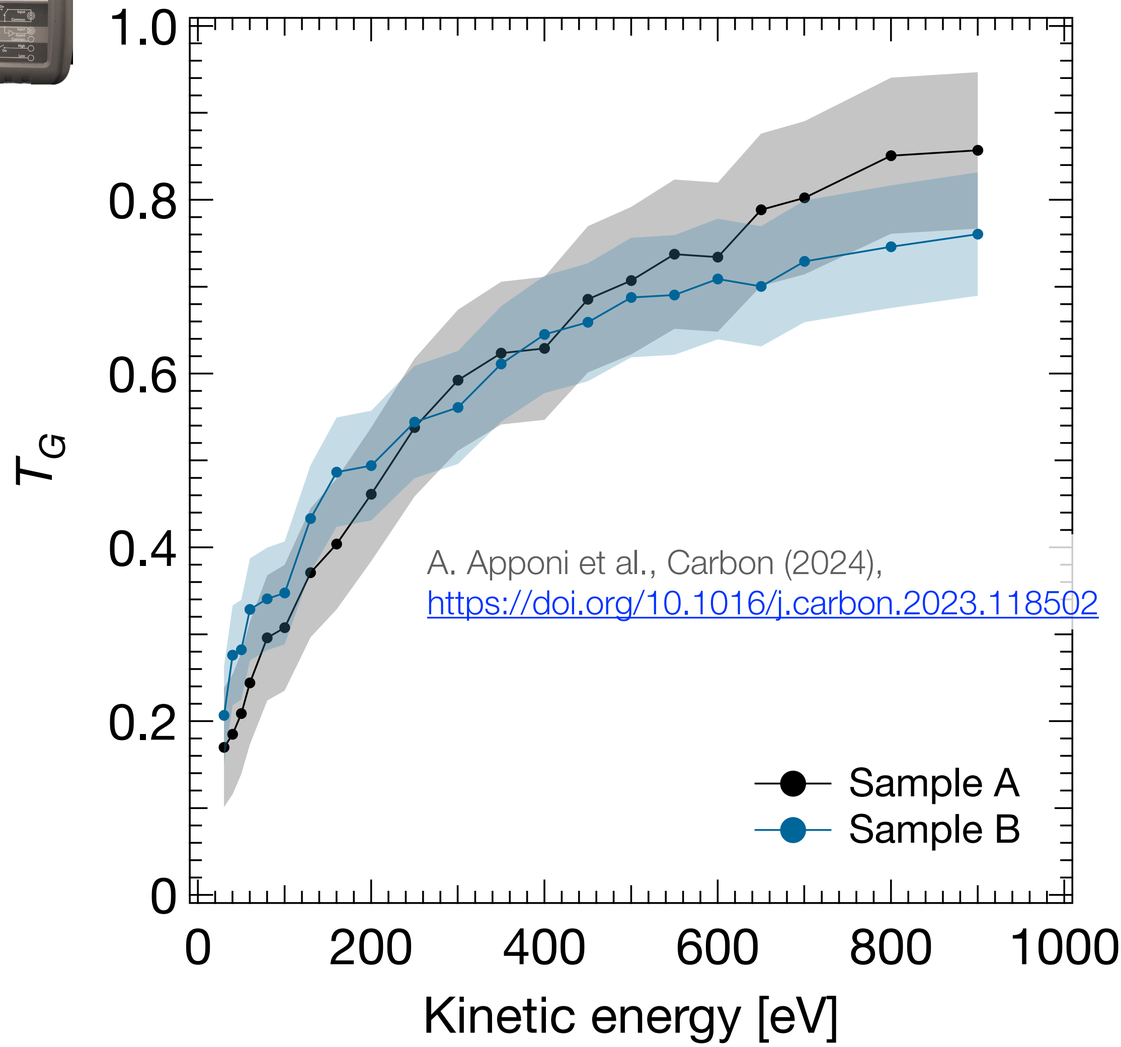


$$I_S(E) = a[I_0 T_{grid} T_G(E)] + b[I_0 T_{grid}]$$

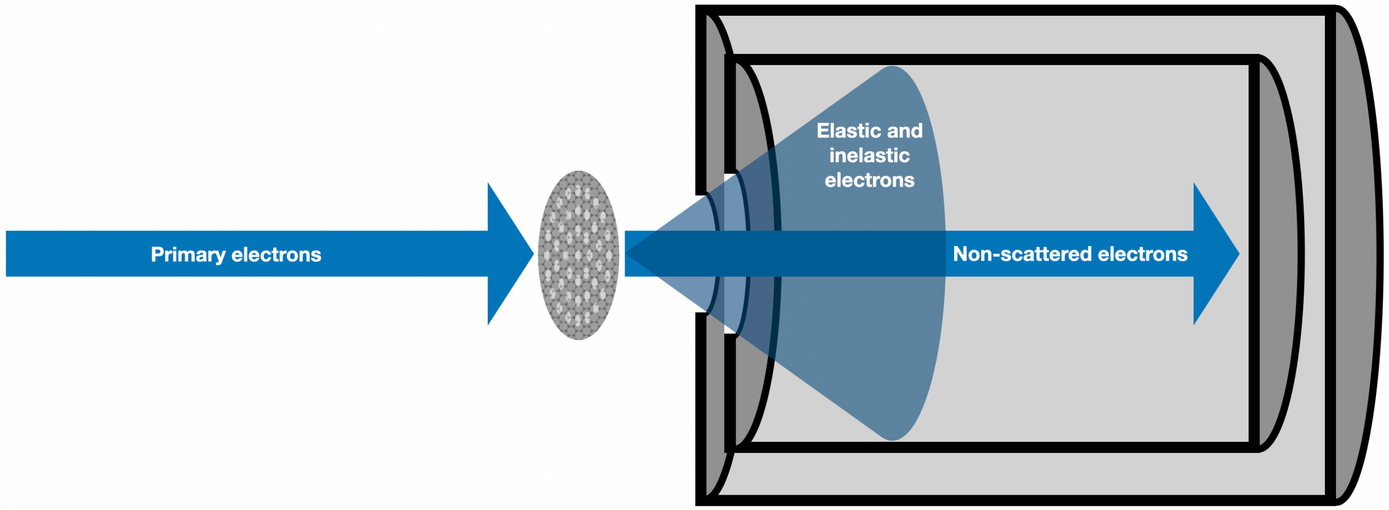
$a + b = 1$

a: Graphene coverage

$$T_G(E) = \frac{\frac{I_S}{I_0 T_{grid}} - b}{a}$$



Energy Resolution for Direct Measurement of Total Cross-Section

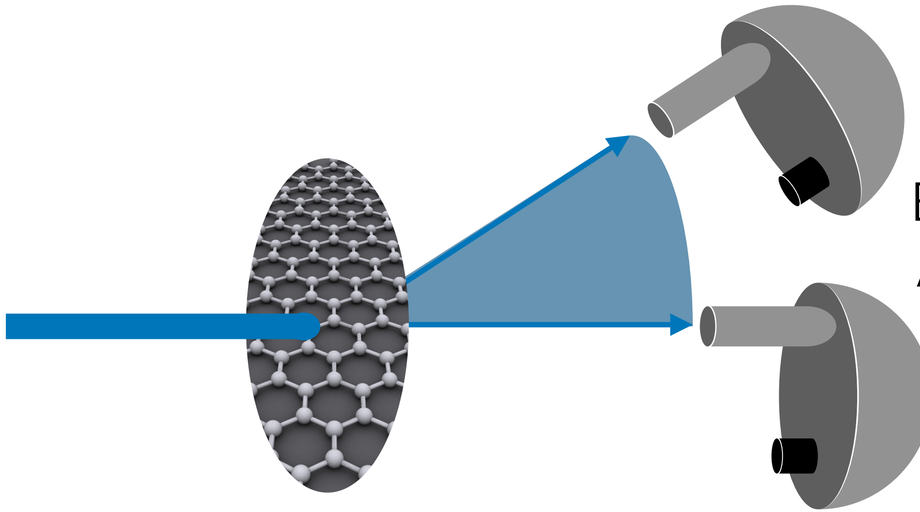


Integral measurement:

- No energy selection
- Current integrated within angular acceptance
- Result setup-dependent

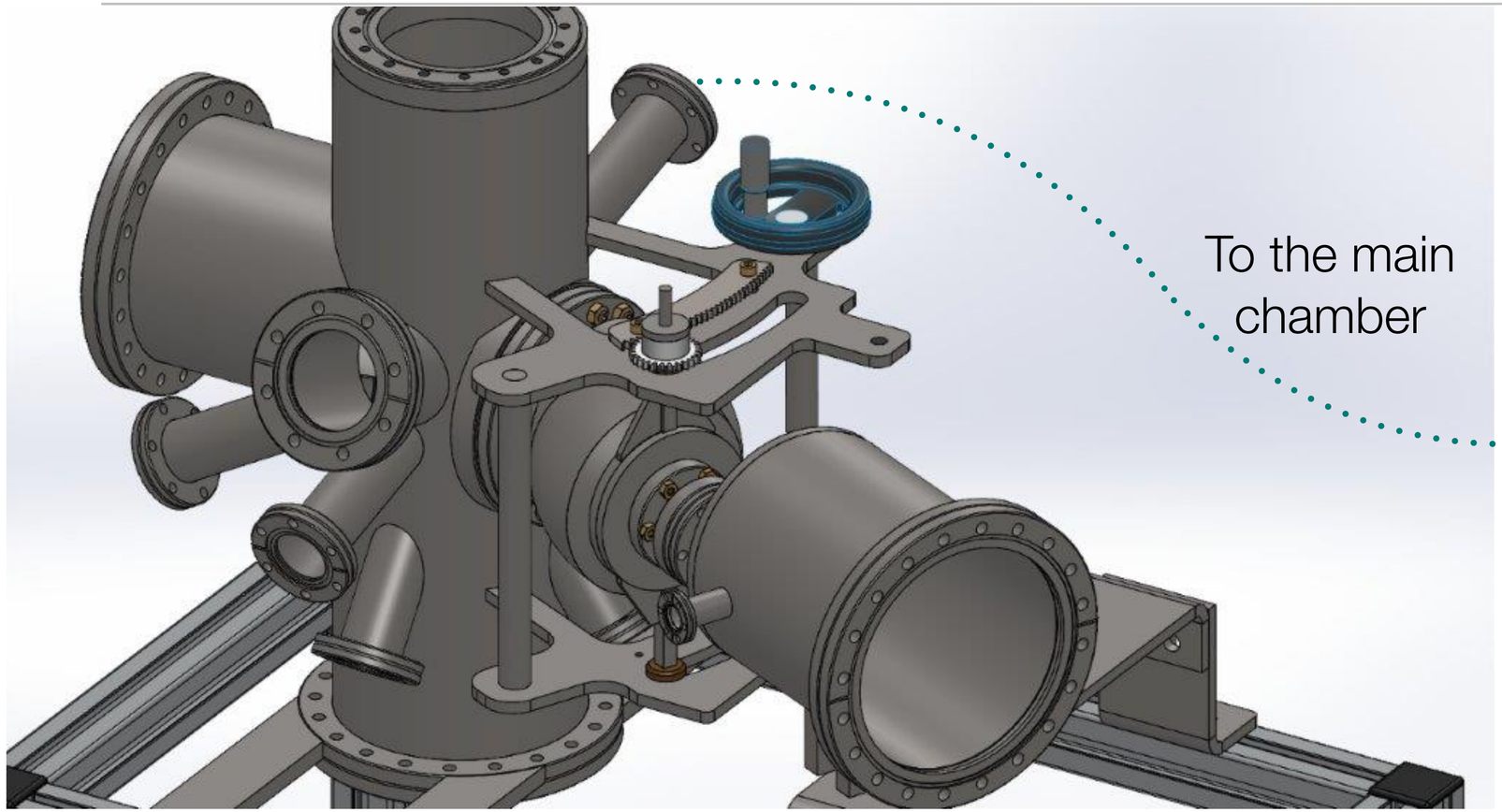
Differential measurement:

- Energy and momentum selection with hemispherical electron analyser
- Direct measurement of σ_{tot} = total cross-section
- Moveable setup for differential cross-section

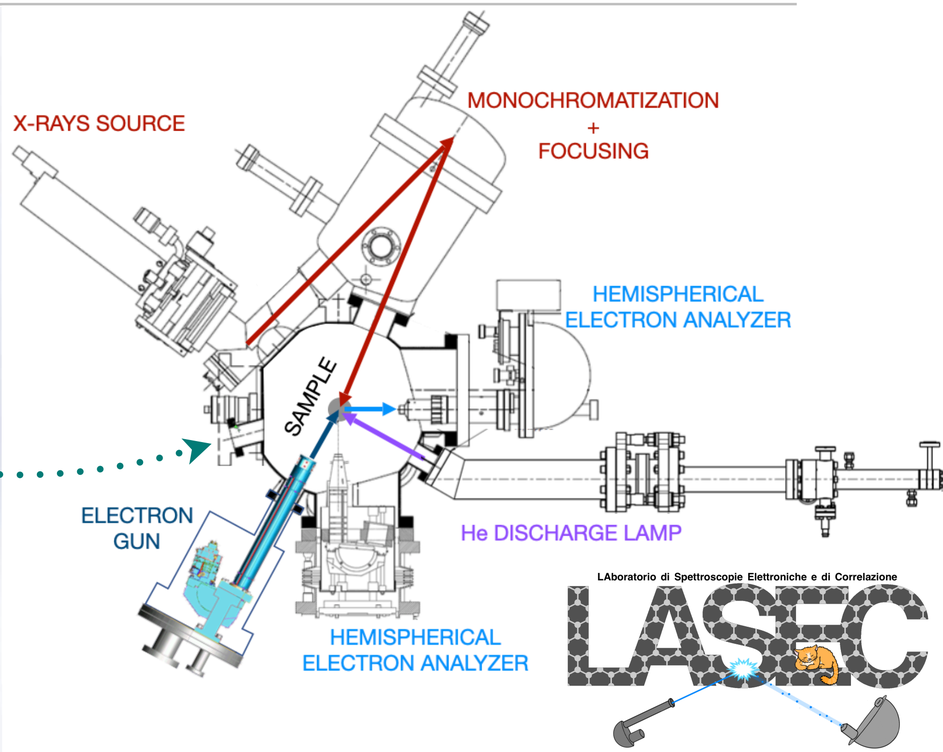


$$I(E) = I_0 e^{-n_G \sigma_{tot}(E)}$$

New Transmission Chamber is Currently Under Contruction



To the main chamber

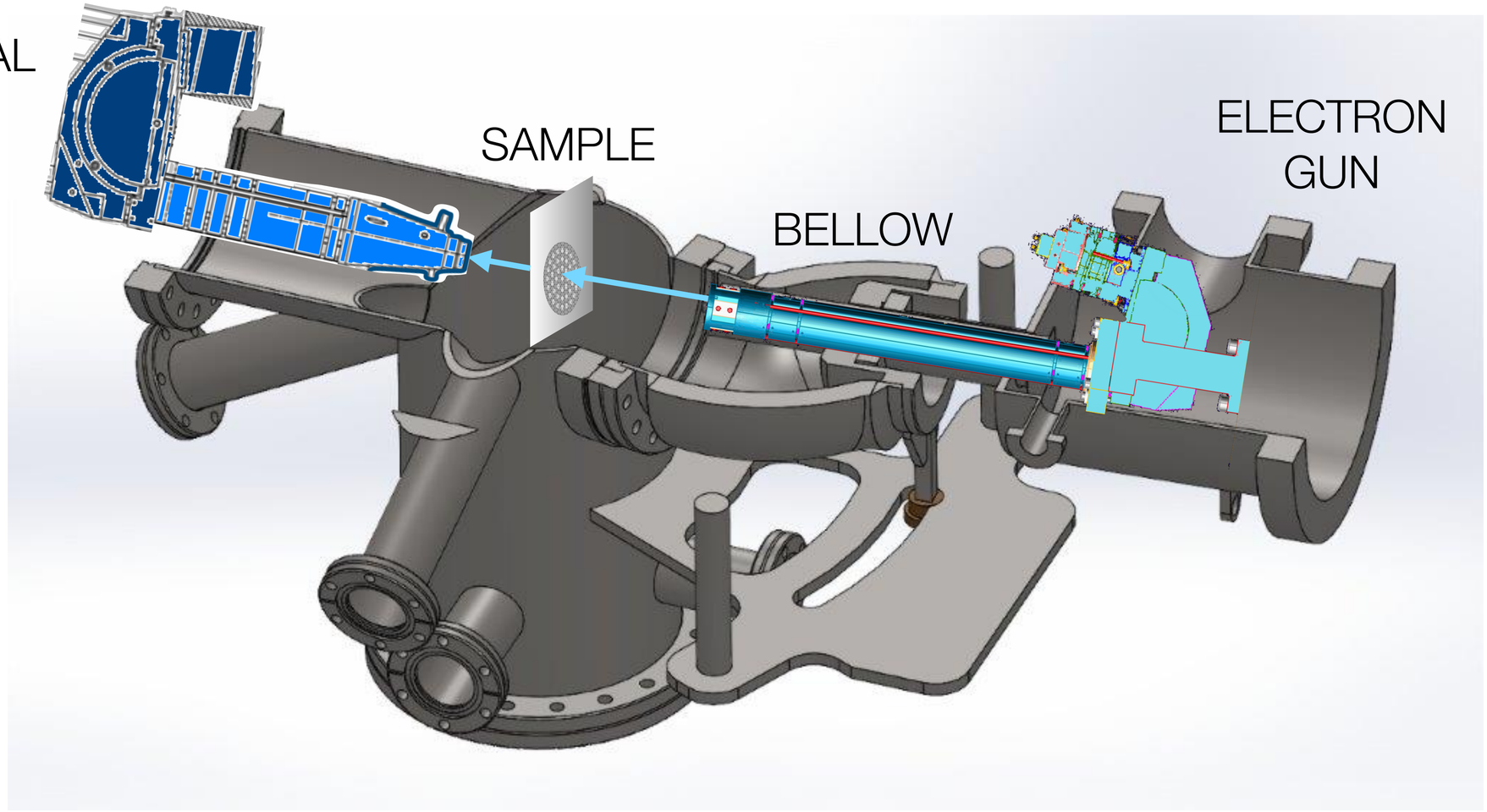


New transmission chamber:

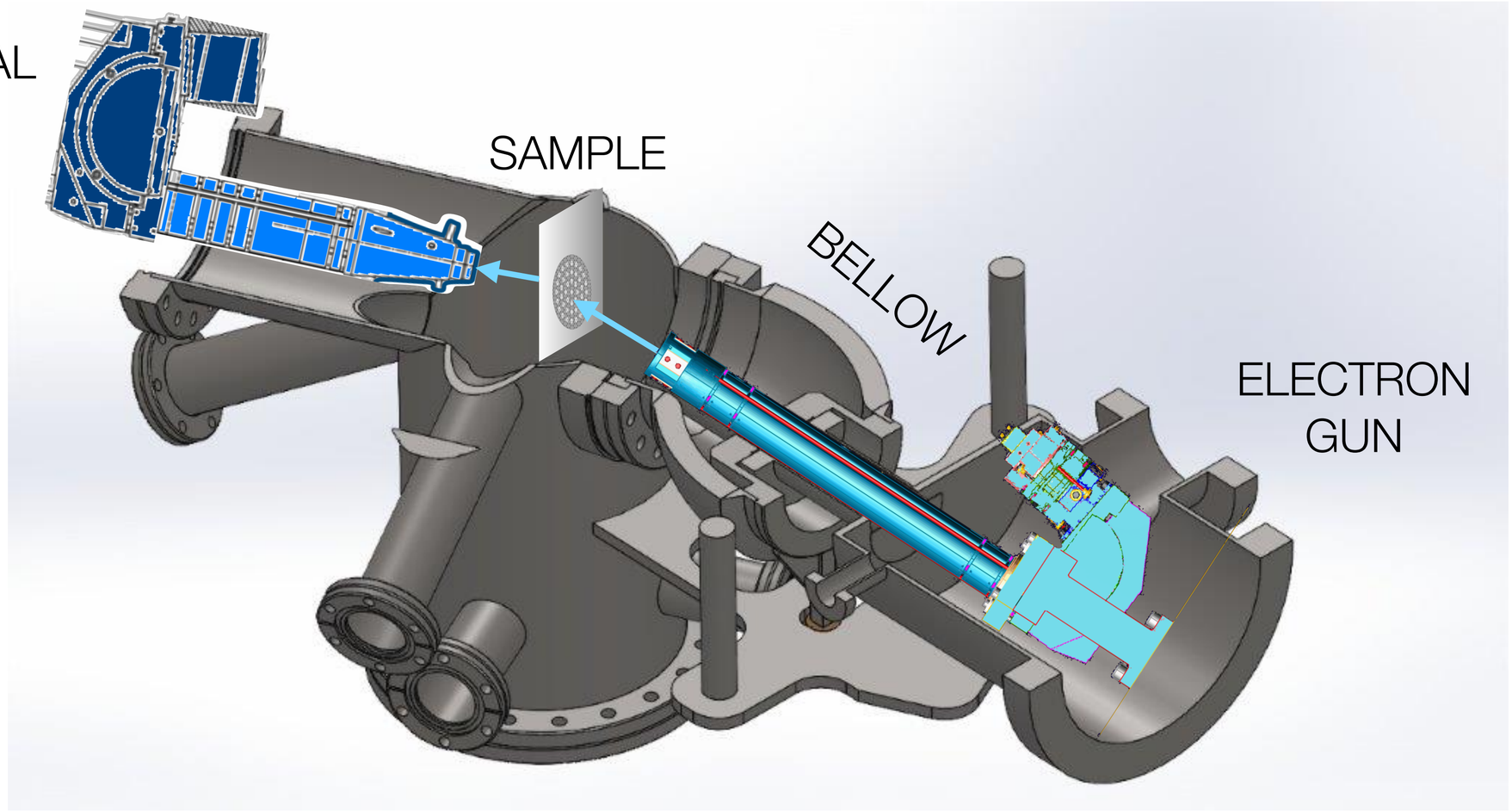
- Currently under construction
- Coupled to the LASEC main chamber in the first place
- Standalone apparatus in the II year

Delivery DECEMBER 2024

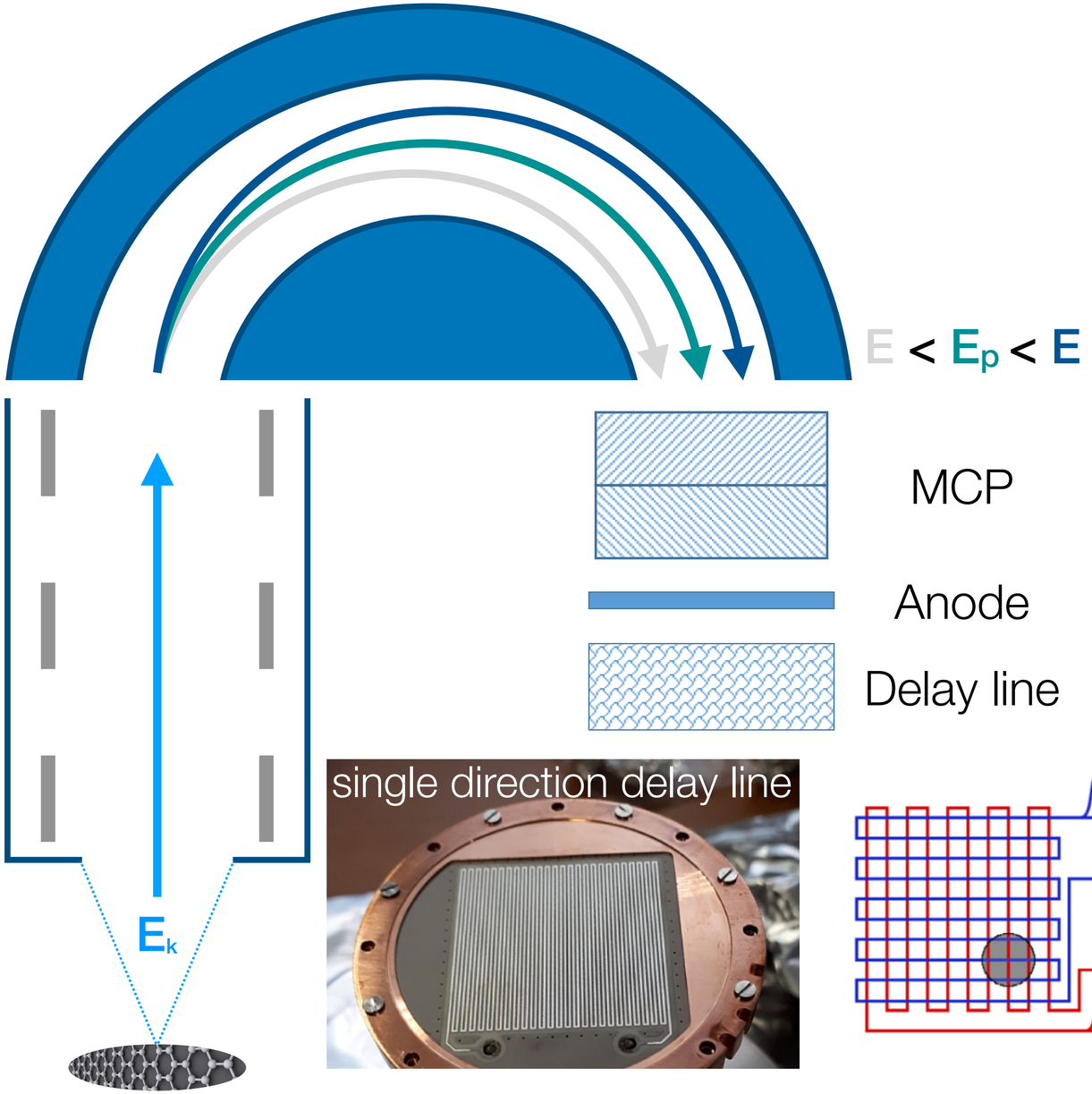
Moveable Electron Source for Angular Dependent Measurement



Moveable Electron Source for Angular Dependent Measurement



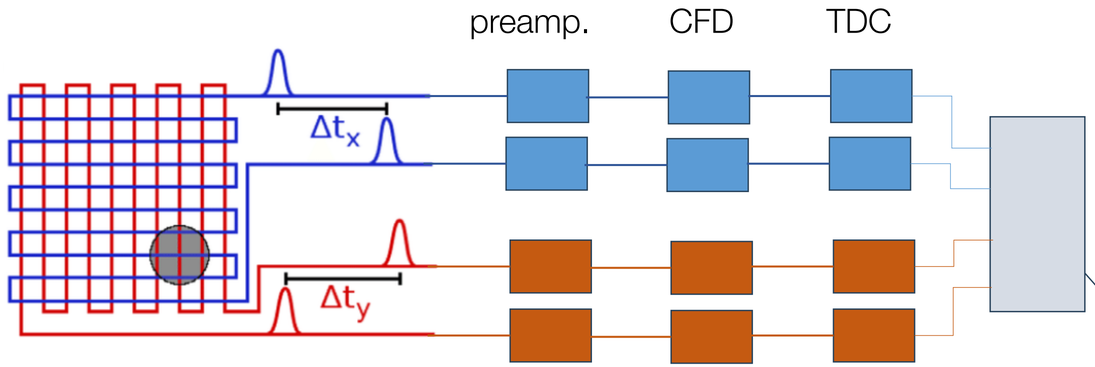
Hemispherical Electron Analyser: an Electrostatic Filter for Electrons

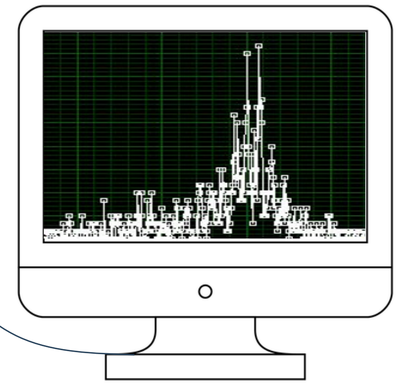


Hemispherical electron analyser:

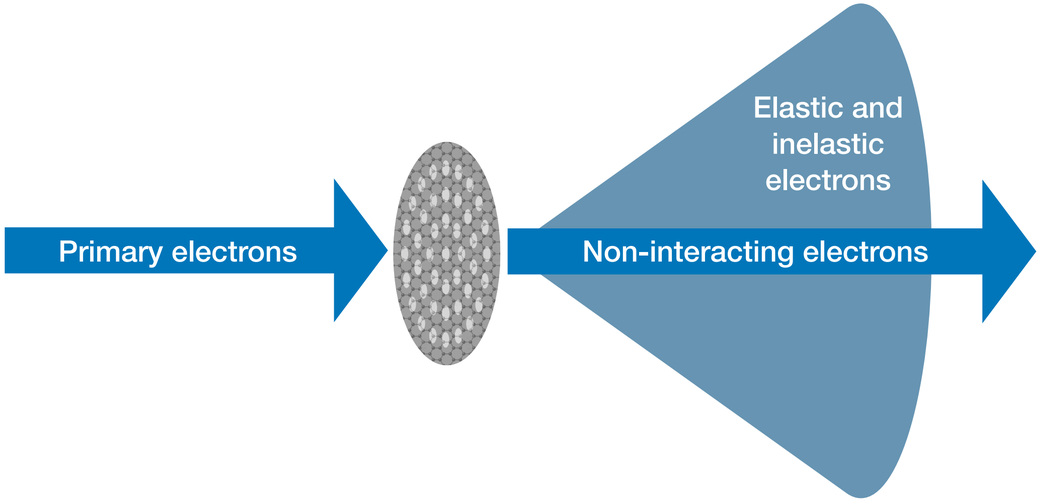
- Electrostatic filter
- Channeltron at exit
- Final electron position ∝ its energy

MCP+Delay line for parallel counting can be implemented





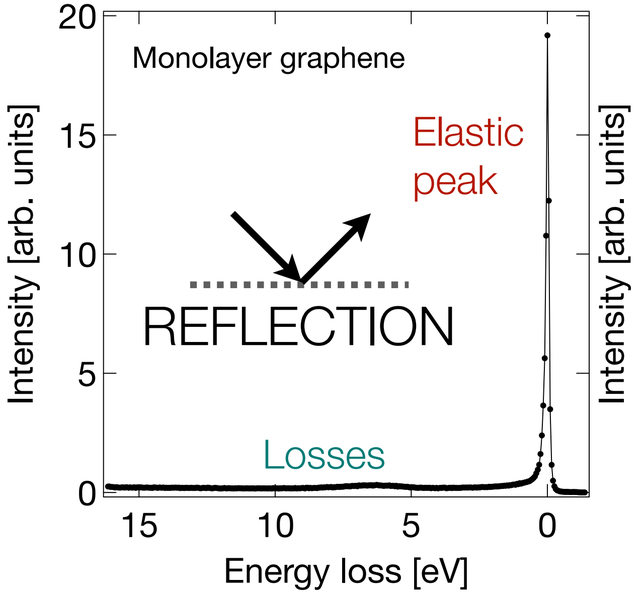
Start with Non-Interacting e-, Boost with Parallel Acqisition for Scattered e-

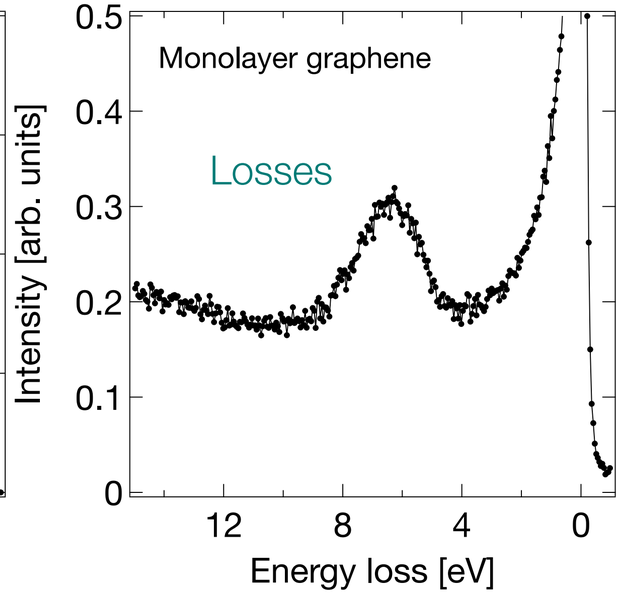


Non-interacting electrons:

- Measure at "0 angle" (1° angular acceptance)
- Total cross-section

I year goal





Scattered electrons:

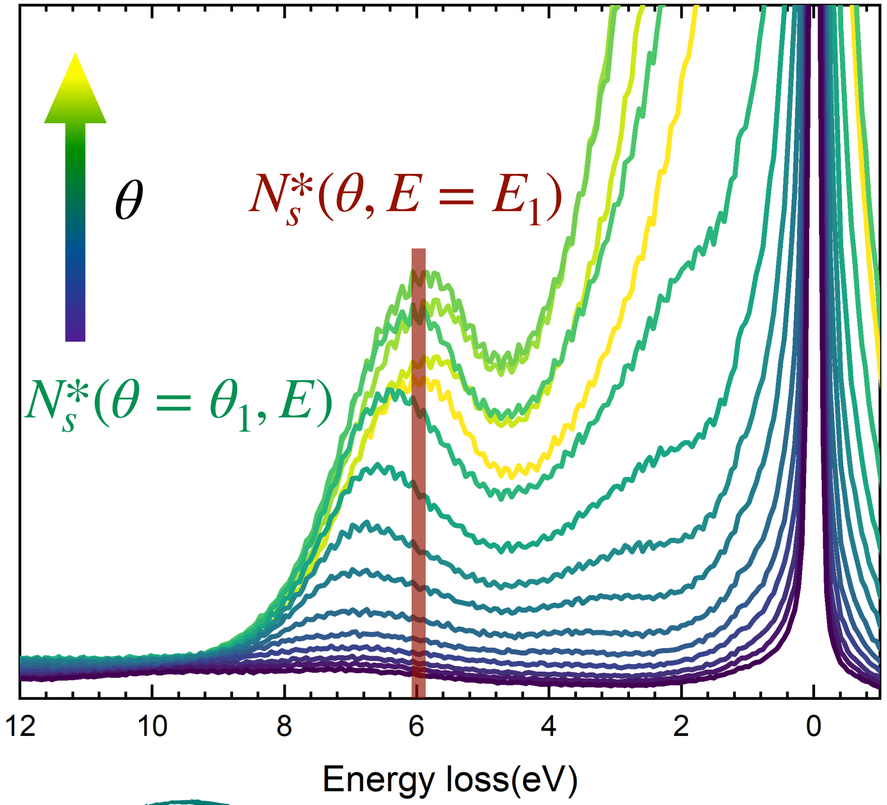
- Reflection spectrum acquired in 30 hours with channeltron
- MCP+delay line for parallel acquisition gives a (*vital*) 10x boost
- Similar acquisition time in transmission
- Differential and total cross-section

II year goal

But How Do You Measure the Cross-Section?

Energy loss measurement:

- Distribution of the transmitted electrons N_s^*
- Spectrum at a given angle —> $N_s^*(\theta = \theta_1, E)$
- Repeat for **different angles**
- Slice at a given energy loss —> $N_s^*(\theta, E = E_1)$
- Repeat for different primary energies



Considering the detector efficiency ϵ

$$N_s^* = N_s\ \epsilon$$

WE CAN MEASURE THIS!

$$\sigma = \frac{N_s}{N_0 n_G}$$

$n_G \sim 39\ nm^{-2}$

A. Apponi et al., Measur.Sci.Tech (2022), 10.1088/1361-6501/ac3d07

The GREEAT Project Needs 145 k€ to Come True

WHAT WE (WILL) HAVE

- Transmission chamber
- Hemispherical electron analyser
- Custom-made electron gun
- Manipulator (5 DoF)
- Drilled sample holder
- Faraday cup
- Commercial electron gun 5 keV

Financial cooperation of:
Dipartimento di Scienze, Università Roma Tre
UTMOST and RD_PTOLEMY, INFN

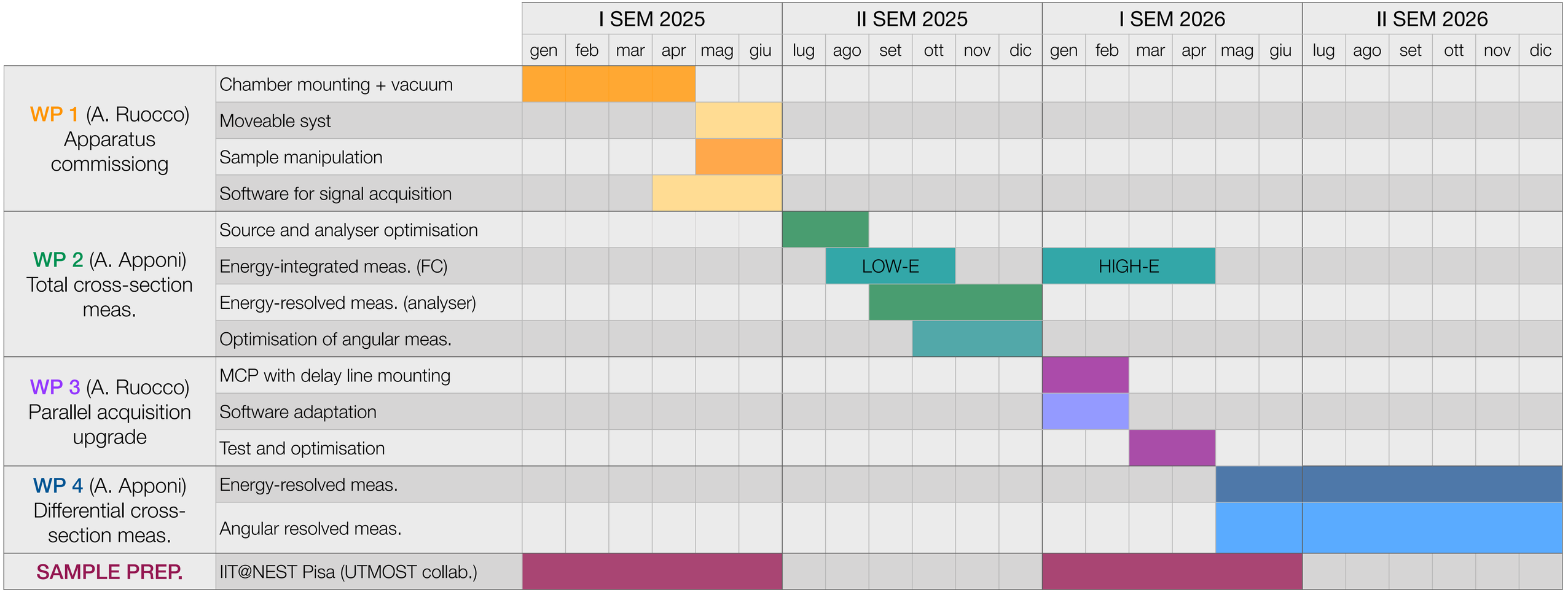
WHAT WE NEED

I year
- Ultra-high vacuum system
- Fast-timing acquisition chain
- Femto/Picoammeter 0.01fA resolution
- Magnetic arm
- Automation

II year
- MCP with delay line for parallel acquisition
- HV floating power supply system

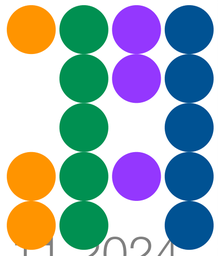
Total request
145 k€

GREEAT Timeline and Team



UNIT 1

UNIT 2



Name	Position	Unit	FTE
Alice Apponi[1,2]	Post-doctoral fellow	INFN Sezione di Roma Tre	0.8
Alessandro Ruocco[2]	Associate professor	Dipartimento di Scienze, Università Roma Tre (associato INFN)	0.4
Daniele Paoloni	Post-doctoral fellow		0.2
To be hired	Post-doctoral fellow		0.2
Orlando Castellano	PhD student		0.2
TOT			1.8

+ 0.2 UTMOST and RD_PTOLEMY

[1]Principal investigator
[2]Unit coordinator

GREEAT project will measure the cross-section of electrons-graphene interaction (and any other 2D!)

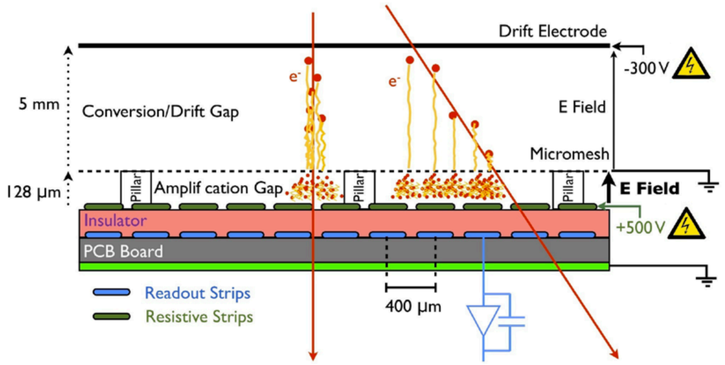


UTMOST

UlTrathin Materials fOr gaSeous deTectors

Design gas detectors with highly improved performances:

- Graphene on the mesh (or GEM)
- Conversion and avalanche gas(es) optimised in mixture and pressure
- Ion back-flow prevented



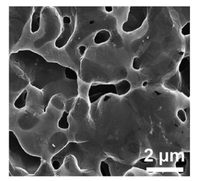
Aroused interest of DRD1 collaboration

CERN

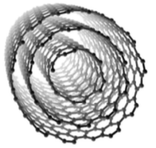
PTOLEMY

Development of novel neutrino detector PTOLEMY:

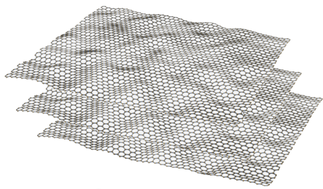
- Graphene-based target
- Transmission of electrons (higher energy) is crucial



Nanoporous graphene



Carbon nanotubes



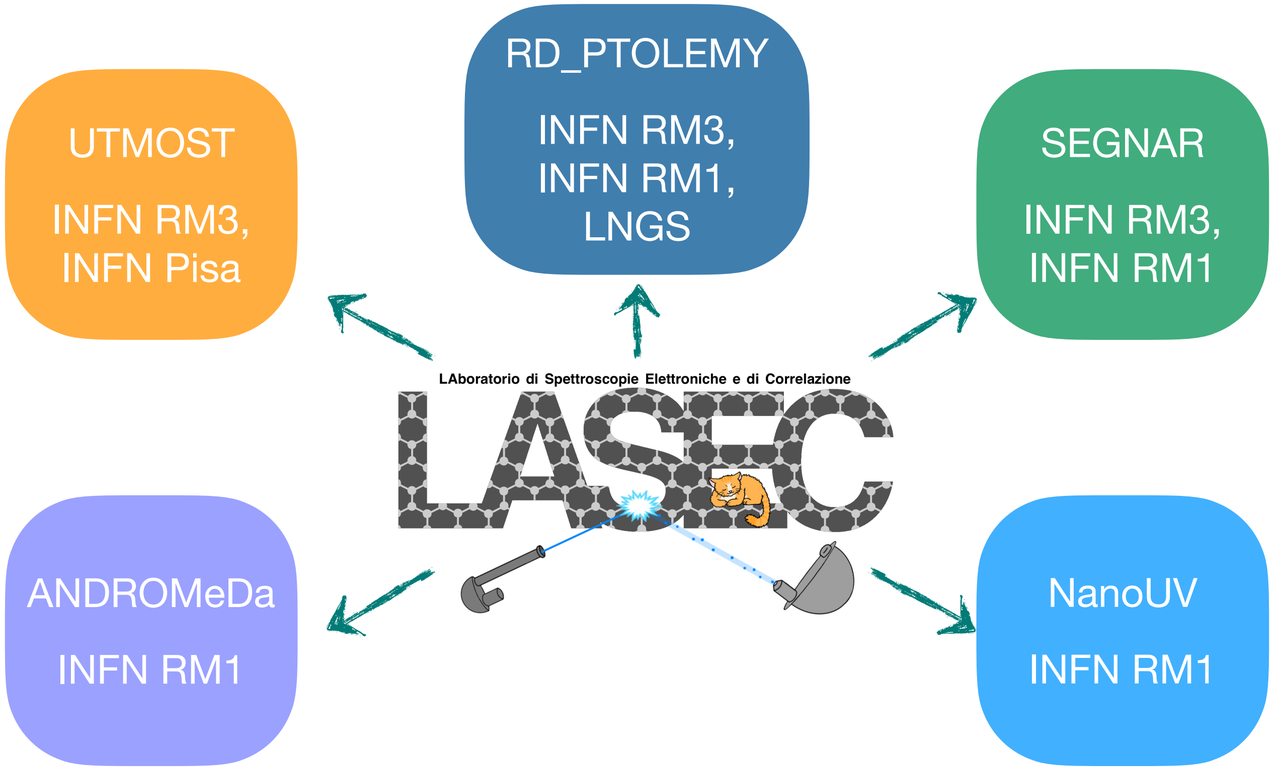
Stacked graphene

To Conclude

The GREEAT project:

- Graphene-electrons cross-section measured for the first time
- Fundamental for applications: MPGD, PTOLEMY

The GREEAT project would reinforce the LASEC lab. (Univeristà Roma Tre) involvement in INFN activities



BACKUP

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LAboratorio di Spettroscopie Elettroniche e di Correlazione

LASEC

UTMOST

INFN RM3,
INFN Pisa

RD_PTOLEMY

INFN RM3,
INFN RM1,
LNGS

SEGNAR

INFN RM3

ANDROMeDa

INFN RM1

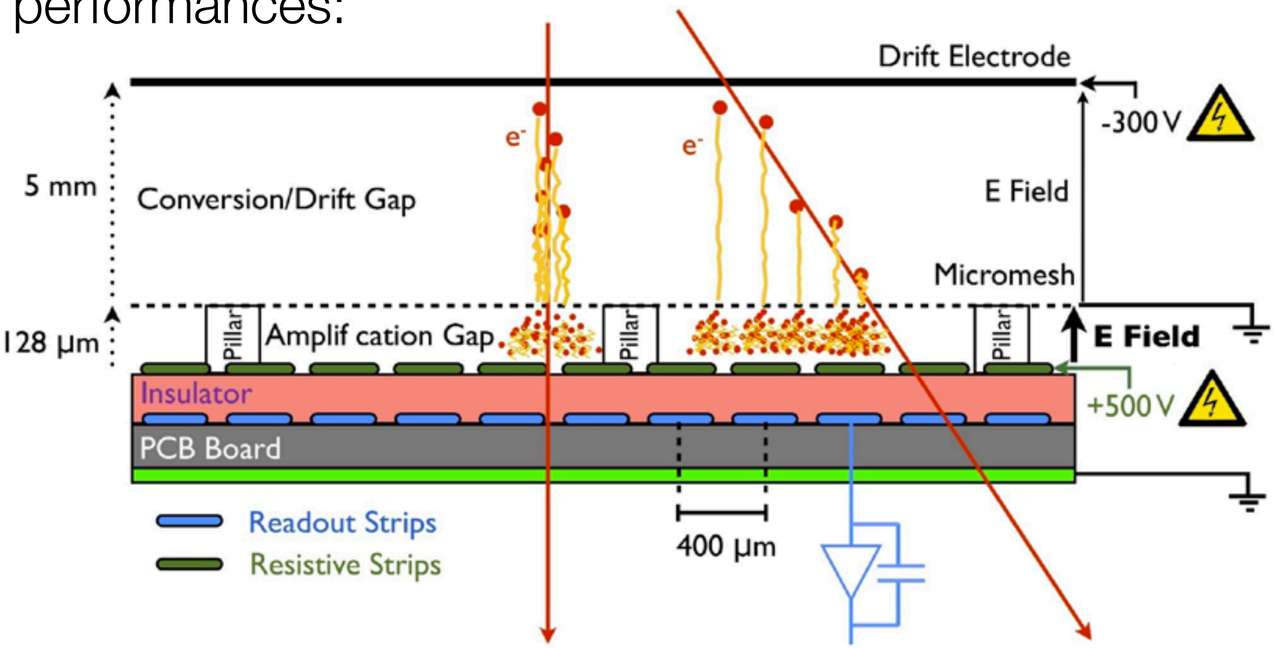
NanoUV

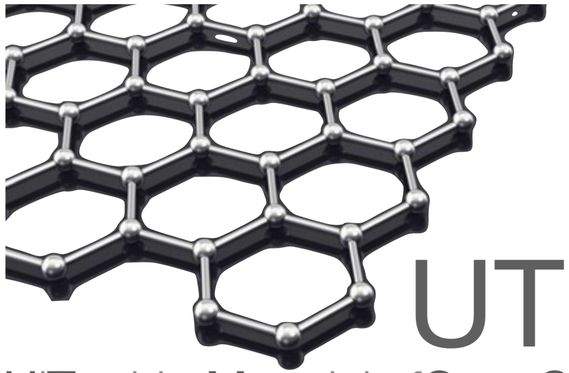
INFN RM1

GREEAT project will measure the cross-section of electrons-graphene interaction (and any other 2D!)

Design gas detectors (MPGD) with highly improved performances:

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- Ion back-flow prevented





UTMOST

UlTrathin Materials fOr gaSeous deTectors

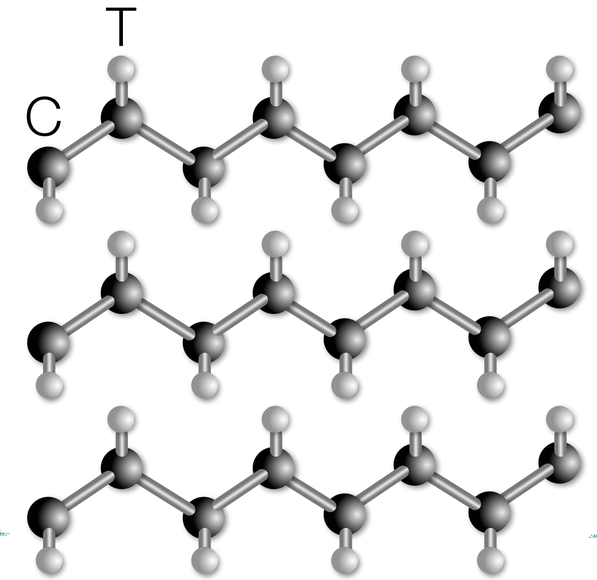
Aroused interest of DRD1 collaboration

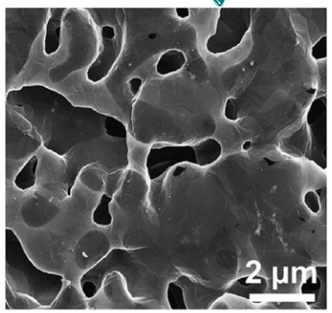
CERN

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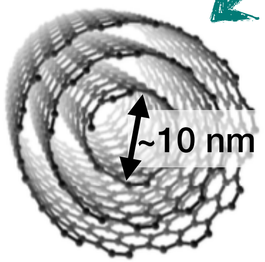
Development of novel neutrino detector PTOLEMY:

- Neutrino mass and cosmic neutrino background
- Measure the β-spectrum at end-point (18 keV)
- Tritium (T) in a graphene-based target
- Wrapped/Stacked configurations to improve density

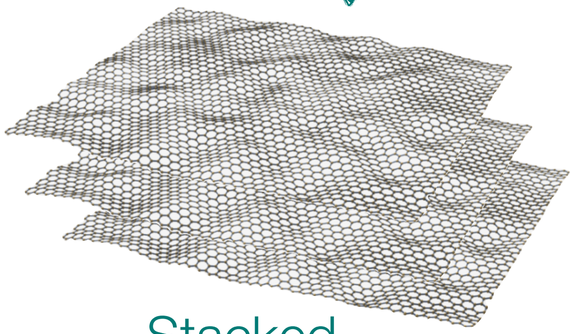




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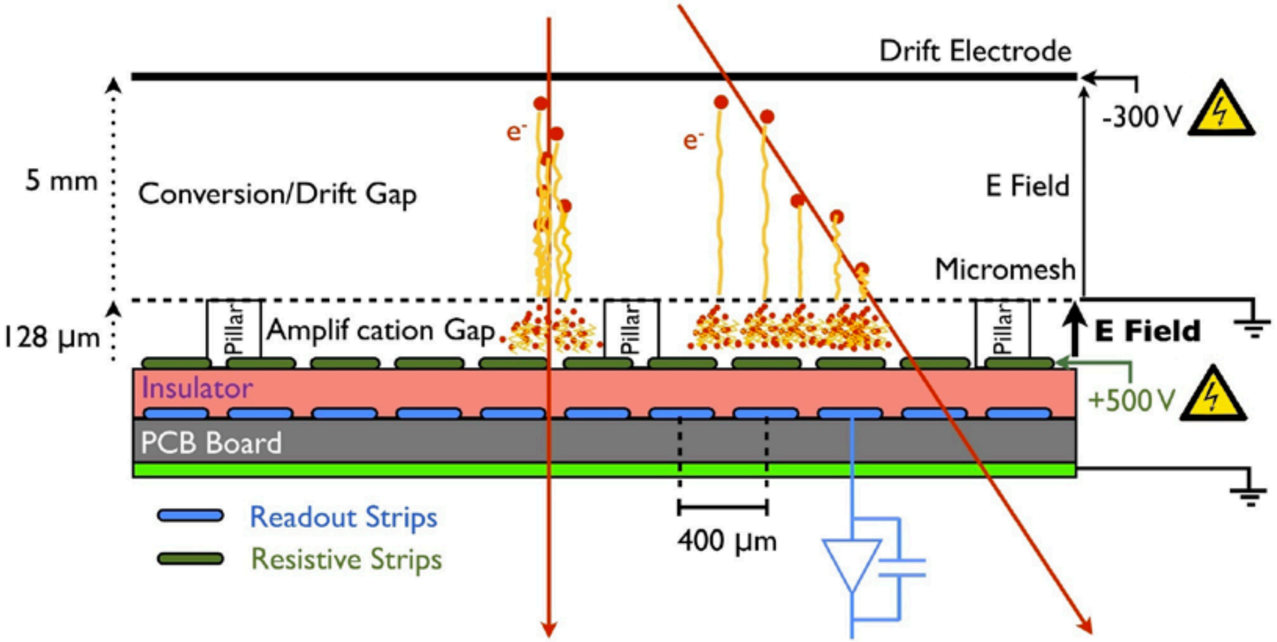


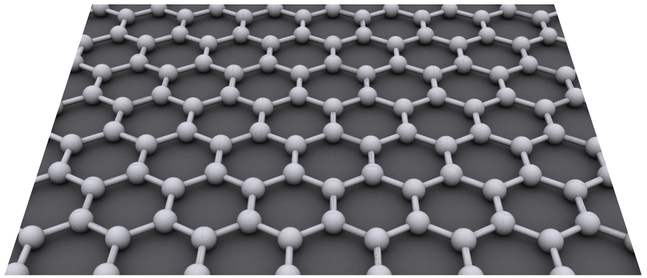
Stacked graphene

Graphene Integration to Improve Detector Performances

Micro Pattern Gaseous Detectors (MPGD):

- Two gaps —> conversion and amplification
- Different gas characteristics to optimise performances
- What if the separation was made of graphene?



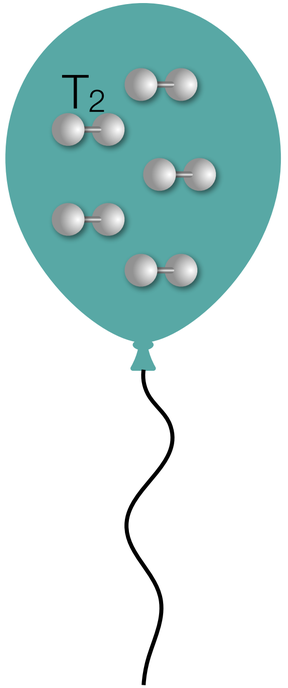


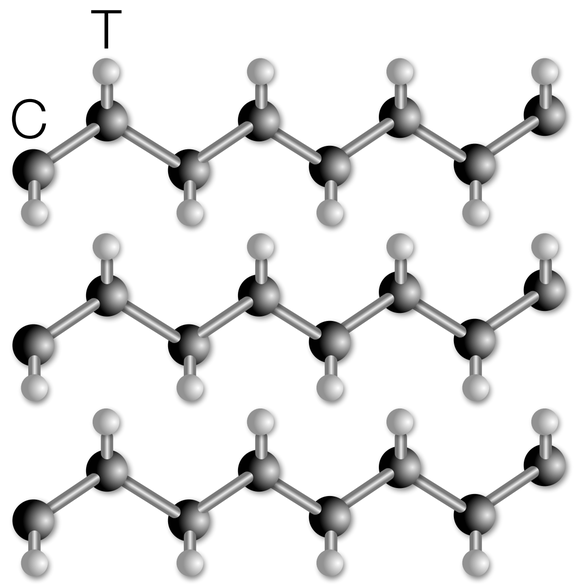
- Atomically thin layer
- Impermeable to atoms
- Transparent to electrons?

T on Graphene as Ptolemy Target

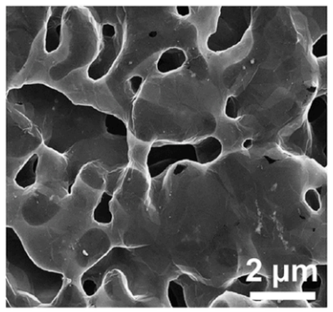
PTOLEMY experiment:

- Cosmic neutrino background and neutrino mass
- Measure the β-spectrum at end-point
- Tritium (T) in a solid-state target
- Carbon nanostructures main candidates
- Wrapped/Stacked configurations to improve density

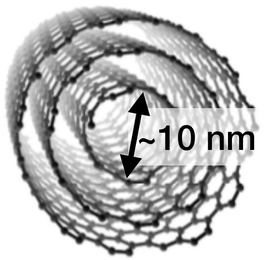




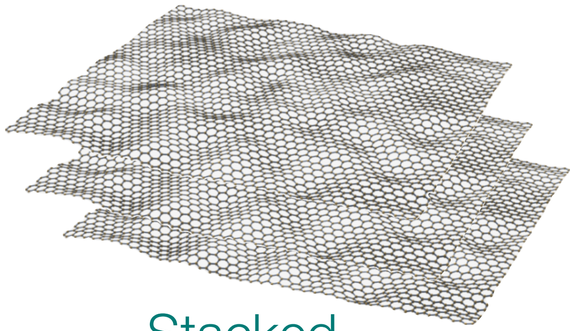
Graphene - electron transmission?



Nanoporous graphene

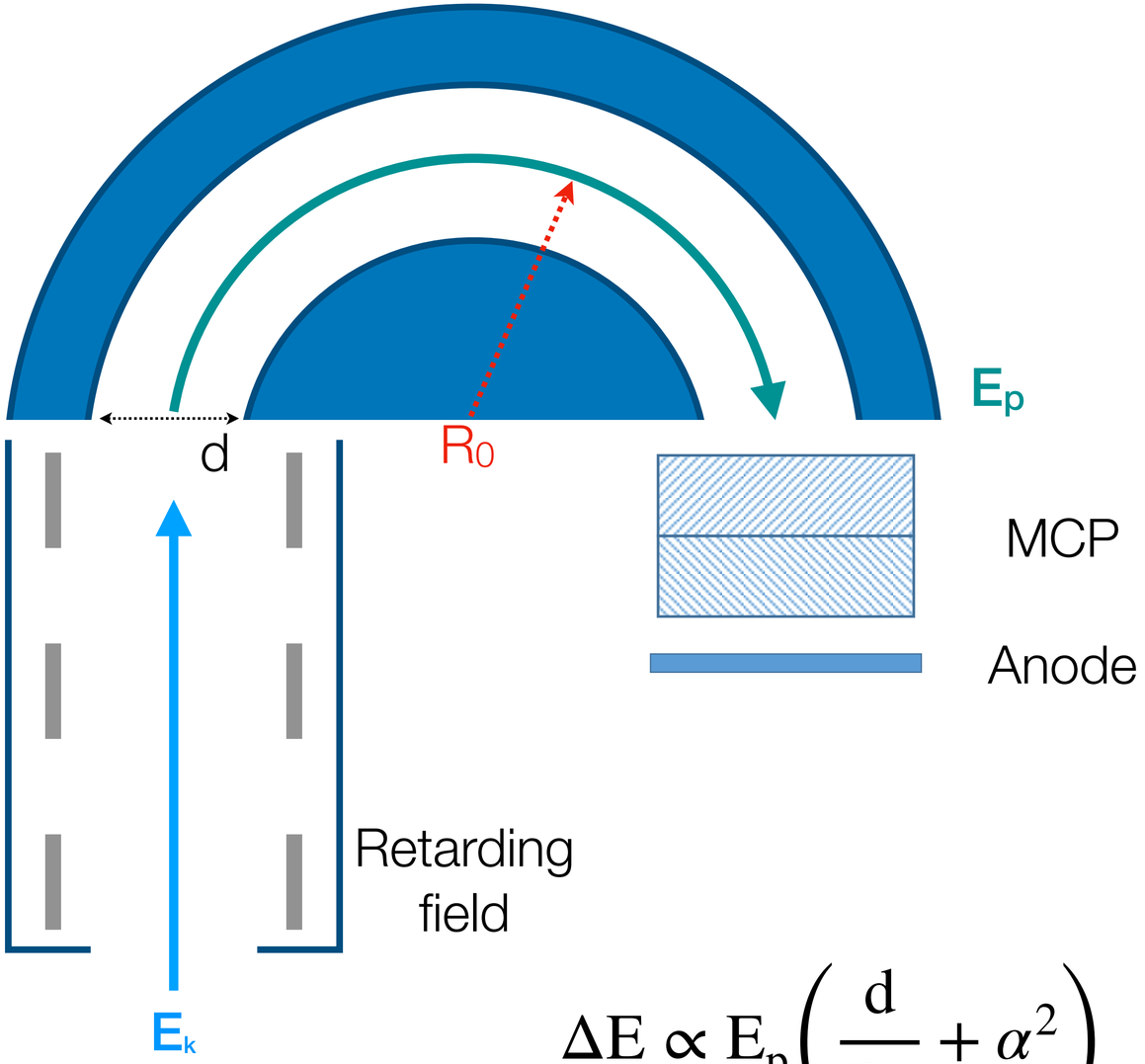


Carbon nanotubes



Stacked graphene

Hemispherical Electron Analyser: an Electrostatic Filter for Electrons

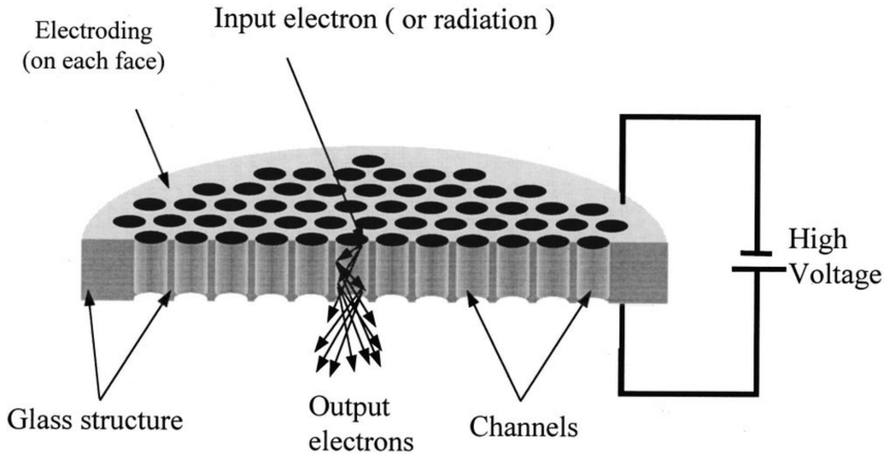


$$\Delta E \propto E_p\left(\frac{d}{R_0}+\alpha^2\right)$$

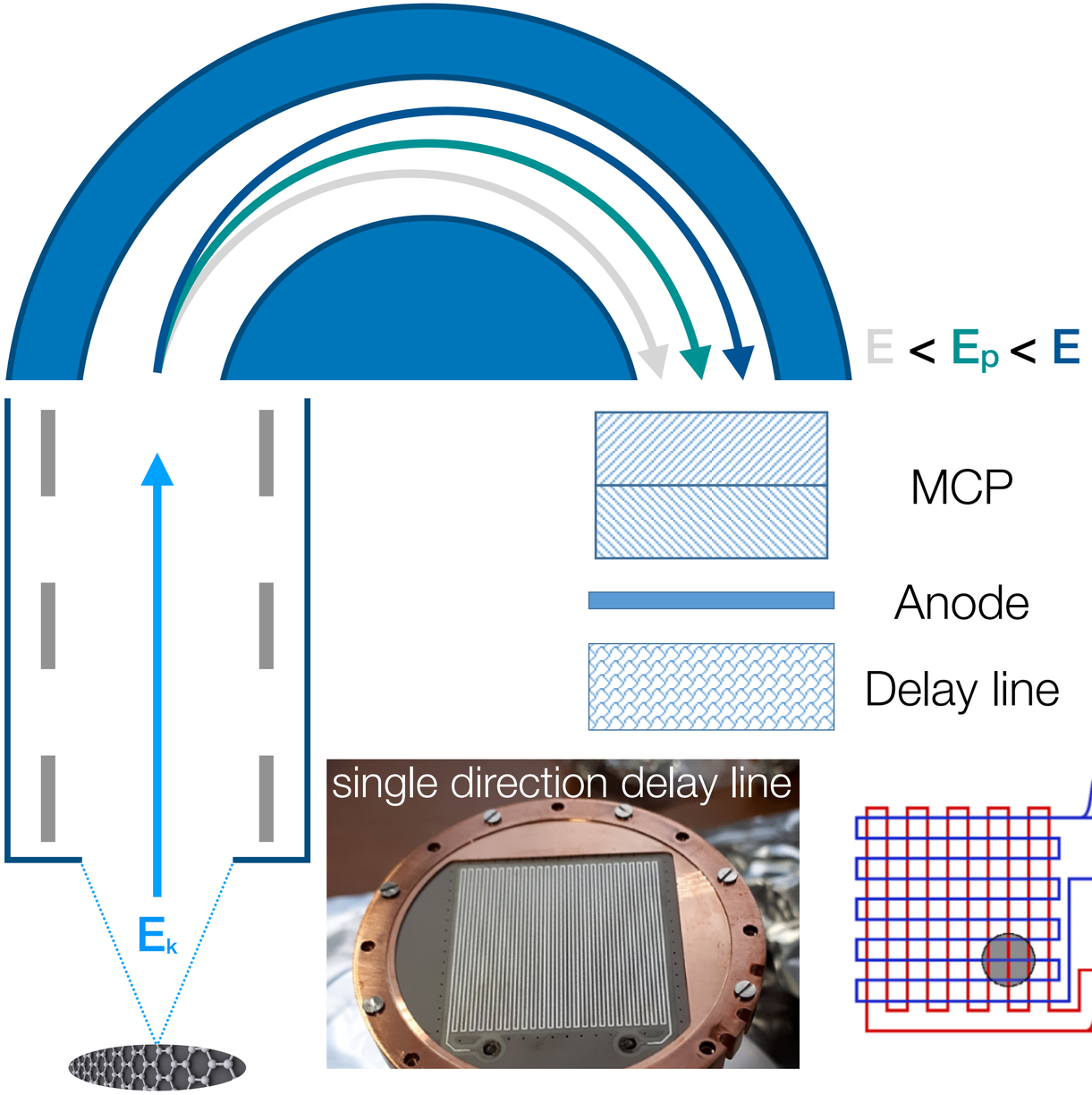
Hemispherical electron analyser:

- Electrostatic filter
- Electron detector at exit

Channeltron / microchannel plate (MCP)



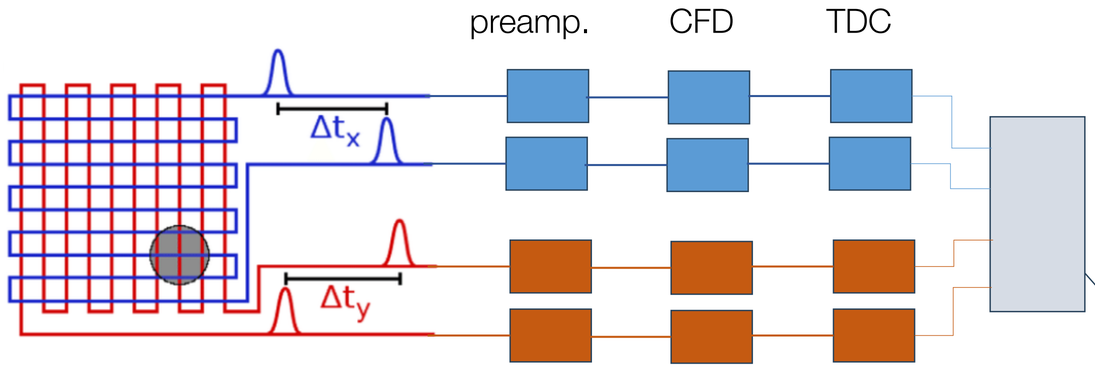
Hemispherical Electron Analyser: an Electrostatic Filter for Electrons

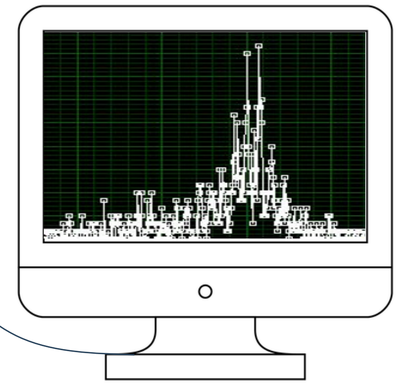


Hemispherical electron analyser:

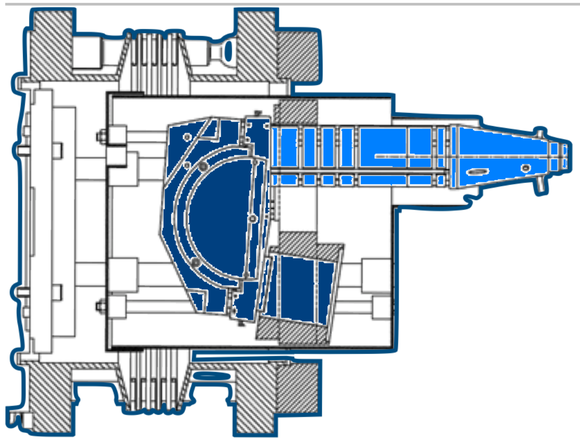
- Electrostatic filter
- Channeltron at exit
- Final electron position ∝ its energy

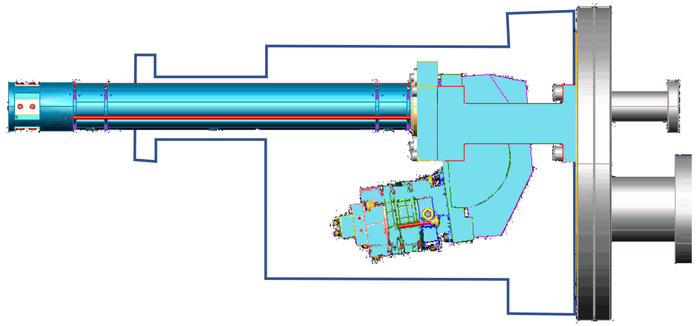
MCP+Delay line for parallel counting can be implemented





Run the Experiment Backwards: Electron Currents of 10 - 100 fA





Electron detector side:

- Max rate ~600 kHz (~100 fA) for linear-mode operation
- Absolute efficiency of MCP measured with LASEC apparatus

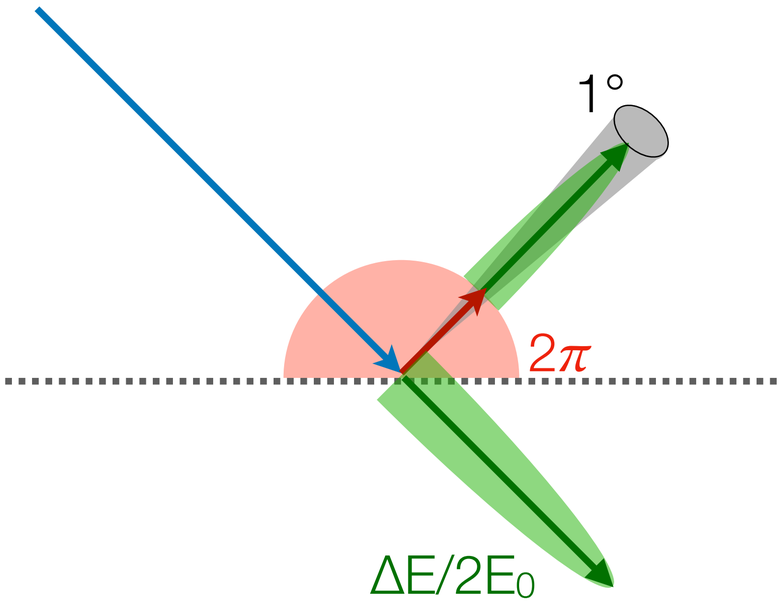
A. Apponi et al., Measur.Sci.Tech (2022), 10.1088/1361-6501/ac3d07

Monochromatic electron gun:

- Continuous beam ~1 fA - ~40 nA
- Tuneable energy within 30 - 900 eV
- Optimal stability even down to a few fA

We will run the experiments with electron currents in the 10 - 100 fA range

Reflection vs Transmission signal



REFLECTION PROCESS: EL+ANEL

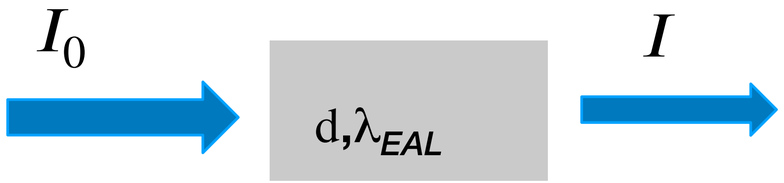
Electrons accepted within analyser

$$\frac{(0.5/57)^2\pi}{2\pi} \sim 4\text{x}10^{-5}$$

TRANSMISSION PROCESS: ANEL

TRANSMISSION = $4\text{x}10^5$ x REFLECTION

Thickness of a 2D material is an ill-defined quantity

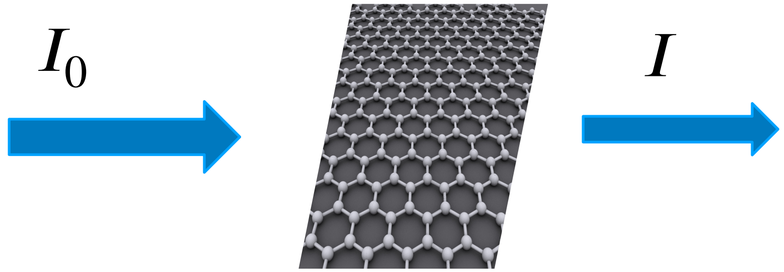


$$I(E) = I_0 e^{-\frac{d}{\lambda_{EAL}}}$$

Which thickness d should be used for ML graphene?

- Interplanar distance in graphite: 3.35 Å
- Twice the radius of covalent bond: 2.48 Å

The attenuation length affected by the arbitrariness of the thickness choice



$$I(E) = I_0 e^{-n_G \sigma_{tot}(E)}$$

- n_G is the surface density of the carbon atom in graphene
- σ_{tot} is the total cross section

A correct measurement of the non scattered electrons I*(E)* allows to obtain the total cross section